What is Electrode?

Electrodes can be defined as conductors that are used to make electrical contact with a non-metallic part of the circuit. The term was first coined by William Whewell and derived from Greek words Elektron, which means "amber" and hoods which translates to "a way". When the current leaves the electrode it is known as the cathode while when the current enters the electrode it is known as an anode.

*****Types of electrodes:** There are different types of electrodes as mentioned below:

1. **Metal - Metal ion electrode:**

   An electrode of this type consists of a metal dipped in a solution containing its own ions.

   Example- Zn/Zn^{2+}, Cu/Cu^{2+} etc.

2. **Metal-insoluble salt electrode:**

   These electrodes consist of a metal is in contact with sparingly soluble of same salt of the same metal dipped in a solution containing the anion of the salt.

   Example-Calomel ($Hg \mid Hg_2Cl_2 \mid Cl^-$) ,

   Silver- Silver chloride electrode ($Ag \mid AgCl \mid Cl^-$)

3. **Gas electrode:**

   Gas electrode consists of a gas is in contact with an inert metal dipped in an ionic solution of the gas molecules.

   Example: Hydrogen Electrode ($Pt|H_2|H^+$)

4. **Ion selective electrode:**

   In ion selective electrode, a membrane is in contact with an ionic solution, with which it can exchange ions.

Example- Glass electrode, solid state electrode and Liquid membrane electrode

5. **Amalgam electrode:**

These electrodes are used in concentration cells where the different quantities of metals are dissolved in mercury to prepare two half-cells. These are connected through solutions of same electrolyte of equal concentration. Such cells are called amalgam cells.

Example: Weston cell.

Cd(Hg), [Cd] = M_1| $CdSO_4$(aq) | Cd(Hg), [Cd] = M_2

6. **Redox electrode:**

The metal or non-metal electrodes are used to study the properties of chemical and biological molecules by redox reactions. These electrodes are also called as working electrodes. These are connected with auxiliary and reference electrodes.

Examples: Gold, platinum, Glassy carbon, graphite etc.

***ION SELECTIVE ELECTRODE

Certain electrodes are sensitive to only certain ions when they are in contact with different kinds of ions. **"Ion selective electrode is the one which selectively responds to a specific ion in a mixture and the potential developed at the electrode is a function of the concentration of that ion in the solution."**

The ion selective electrode generally consists of a membrane which is capable of exchanging the specific ions present in the solution. Therefore, these electrodes are also called as **membrane electrodes.**

Example: **Glass electrode**, solid state electrode and Liquid membrane electrode

1. **Glass electrode:** **Glass electrode** is an ion selective electrode which is sensitive to the H^+ ions in the solution and hence is widely used for pH determination.

a. Construction of glass electrode:

The glass electrode consists of a long glass tube with a thin-walled bulb at one end. The glass bulb made up of special type of glass of composition Na_2O-22%, CaO-6%, and SiO_2-72%. It has low melting point and relatively high electrical conductivity. The glass bulb is filled with a 0.1 M HCl solution and insert Ag/AgCl electrode, which acts as internal reference electrode and connected by platinum wire for external contact.

The electrode representation is, **Ag|AgCl | HCl (0.1M) |glass**

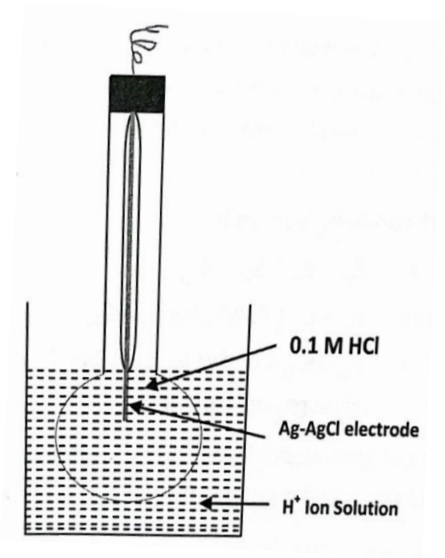
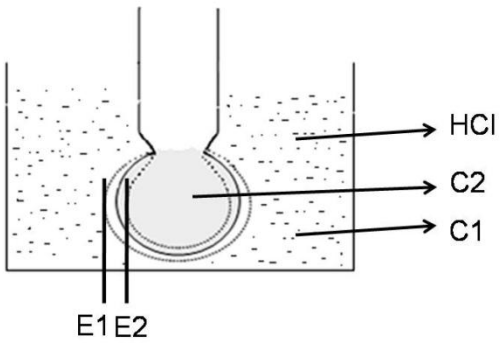


Fig. Glass electrode

b. Working:

When glass electrode is dipped into the analyte, a potential is developed across the glass membrane. This is called boundary potential E_b. The potential is due to the difference in potential ($E_b = E_1 - E_2$) developed across the gel layer of the glass membrane between the two liquids.



Consider, C1 > C2

Where C_1 and C_2 are the concentrations of acids **outside** and **inside** of the glass bulb respectively

E_b related to the difference in the hydrogen ion concentration of the two solutions,

i.e **$E_b = E_1 – E_2$** **(1)**

Apply Nernst equation for equation (1),

$$E_b = \frac{2.303RT}{nF}\log C_1 - \frac{2.303RT}{nF}\log C_2 \qquad (2)$$

E_b = 0.0591 log C1 - 0.0591 log C2

Where, [R= 8.314 J/mol/k, F=96500 C/mol, n=1 for H^+ ion.]

E_b = K + 0.0591 log C1 [where K = -0.0591 log C2] (3)

Where**, K** is a constant which depends on the pH of the solution. Since $C_1 = [H^+]$ of the solution and $-\log [H^+] = pH$, then the equation becomes,

E_b = K - 0.0591pH (4)

This equation (4) is used to determine the pH of the solution.

When the glass electrode is dipped in the test solution, the Na^+ ions of the glass membrane are exchanged for H^+ ions of the test solution.

$$H^+ + Na^+Gl^- \rightleftharpoons Na^+ + H^+Gl^-$$

The boundary potential established due to the above reaction is mainly responsible for the glass electrode potential E_G. The potential of the glass electrode is given by

$$E_G = E_b + E_{\text{Ag/AgCl}}$$

A small potential developed is observed that even when $C_1 = C_2$ is called asymmetric potential (E_{asy}). Then the equation is written as

$$E_G = E_b + E_{\text{Ag/AgCl}} + E_{asy}$$

Substituting E_b, (equation 3) then

$$\mathbf{E_G = K - 0.0591\ pH + E_{Ag\,|\,AgCl} + E_{asy}}$$

$E_G = E^0_G$ **– 0.0591 pH [**where E^0_G is constant i.e, $E^0_G = K + E_{\text{Ag/AgCl}} + E_{asy}$]

Where, E^0_G = Standard emf of glass electrode

On simplifying, pH

$$\mathbf{pH = \frac{E^0_G - E_G}{0.0591}}$$

c. Applications of Glass electrode:

1. pH measurement.
2. Food Analysis.
3. Cosmetic analysis.
4. Environmental regulations.
5. Soil acidity determination.

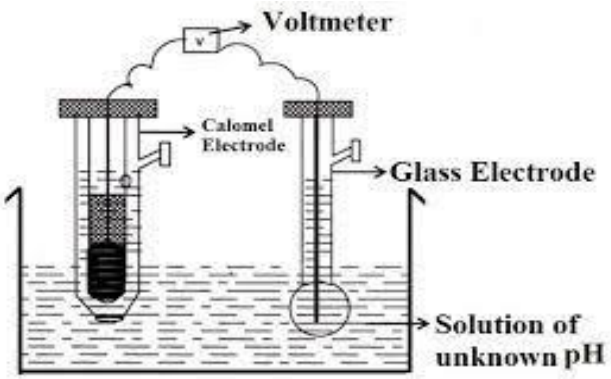
d. Advantages

1. It can be used in presence of oxidizing or reducing substances and metal ions.
2. It is not poisoned easily.
3. It can be used in un-buffered solutions and adopted for measurements with very small quantities of solutions.
4. It can be used in portable instruments.
5. It is simple to operate, hence extensively used in various laboratories.
6. It can be used for colored, turbid and colloidal solutions.

e. Disadvantages

1. The glass membrane though it is very thin, it offers high resistance. Therefore, ordinary potentiometers cannot be used; hence it is necessary to use sensitive potentiometers.
2. It does not function satisfactorily in pure alcohol.
3. In acidic solutions of pH>1, the salt effect due to anions are observed.

One of the application of glass electrode: Determination of pH using glass electrode:



Determination of pH using glass electrode

A glass electrode is coupled with a saturated calomel electrode (SCE), to determine the pH of an acidic solution.

The cell assembly is represented as,

Hg | Hg_2Cl_2 |Cl^- || Solution of unknown pH | glass | HCl (0.1M) | AgCl | Ag

The potential of the above cell, E_{cell} is measured using an electronic voltmeter with a pH meter. The emf of the cell is given by

$$E_{cell} = E_{cathode} - E_{anode}$$

$$E_{cell} = E_G - E_{SCE} \qquad (1)$$

We know that E_G of the glass electrode depends on the pH of the solution in which the glass electrode is dipped. Substitute E_G in above equation (1), thus

$$E_{cell} = E^0_G - 0.0591\ pH - E_{SCE}$$

$$pH = \boxed{\frac{E^0_G - E_{SCE} - E_{cell}}{0.0591}}$$

E^0 value can be determined by dipping the glass electrode in a solution of known pH (buffer solution) and measuring the emf of the cell formed when combined with calomel electrode

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2. Reference electrode:

Reference electrode is metal-insoluble metal salt electrode, are the electrode with reference to those, the electrode potential of any electrode can be measured.

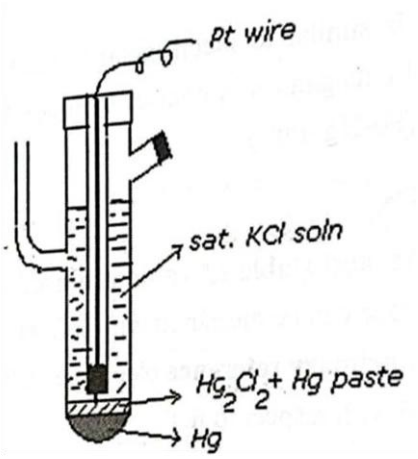
Reference electrode is a reversible electrode of known electrode potential, which is used to determine the potential of the other electrode. Example: SHE electrode and **calomel electrode (Hg | Hg_2Cl_2 | Cl^-).**

Role of reference electrode:

1. The reference electrode maintains a consistent potential regardless of the solution's composition.
2. The voltage difference between the reference and measuring electrodes varies with the pH of the solution.
3. Silver/silver chloride (Ag/AgCl) and Mercury/Calomel (Hg_2Cl_2) are two common reference systems. The Silver/Silver Chloride reference is by far the most commonly used because of its universal applicability, wide operating temperature range, and low toxicity.

Calomel electrode:

a. Construction:



Calomel electrode

Calomel electrode is a metal-insoluble salt electrode. A small amount of mercury is placed at the bottom of a glass tube. It is covered with a paste of mercury and solid mercurous chloride. The two layers of mercury and the paste are covered with a saturated solution of potassium chloride. A glass tube carrying a platinum electrode is inserted in the arrangement to make electrical connections. Also, a salt bridge is immersed in potassium chloride solution to couple calomel with other electrodes. This electrode can be used as

anode as well as cathode to determine the electrode potential of the other electrodes. The calomel electrode can act as anode or cathode depending on the nature of the other electrode of the cell.

Calomel electrode can be represented as, Hg| Hg_2Cl_2|KCl (sat.)

Working

When it act as anode:

$$2Hg \rightarrow Hg_2^{2+} + 2e^-$$
$$Hg_2^{2+} + 2Cl^- \rightarrow Hg_2Cl_2$$

Overall reaction: $2Hg + 2\,Cl^- \rightarrow Hg_2Cl_2 + 2e^-$

When it act as cathode:

$$Hg_2^{2+} + 2e^- \rightarrow 2Hg$$
$$Hg_2Cl_2 \rightarrow Hg_2^{2+} + 2\,Cl^-$$

Overall reaction: $Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2\,Cl^-$

The net reversible reaction is $Hg_2Cl_2 + 2e^- \leftrightarrow 2Hg(l) + 2Cl^-$

The electrode potential of calomel electrode is depending upon the concentration of KCl solution.

The electrode potential of calomel electrode is,

Acc. To Nernst equation, the potential at 298K

$$E_{cell} = E^0{}_{cell} - \frac{0.0591}{n} \log\frac{[P]}{[R]}$$

$$E_{cal} = E^0{}_{cal} - \frac{0.0591}{2} \log\frac{[Hg]^2[Cl^-]^2}{[Hg2Cl2]}$$ Where n= 2

$[Hg] = 1$

$[Hg2Cl2] = 1$

$$E_{cal} = E^0{}_{cal} - \frac{0.0591}{2} \log[Cl^-]^2$$

$$E_{cal} = E^0_{cal} - 0.0591 \log [Cl^-]$$

Note: The potential value of calomel electrode of diff. conc. of KCl measured using SHE and they are ; 0.1 N KCl = 0.3338 V, 1.0 N KCl = 0.282 V, Saturated KCl = 0.2412 V

b. Applications:

- The SCE is used in pH measurement,
- Cyclic voltmmetry and general aqueous electrochemistry

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3. Concentration Cell

A concentration cell is an electrochemical cell in which both the anode and the cathode are made up of the same metal in contact with solutions of the same electrolyte, but of different concentrations. Example: A cell in which both the electrodes are made of zinc metal dipped in $ZnSO_4$ solution of different concentrations as shown in figure.

a. Construction of Cells

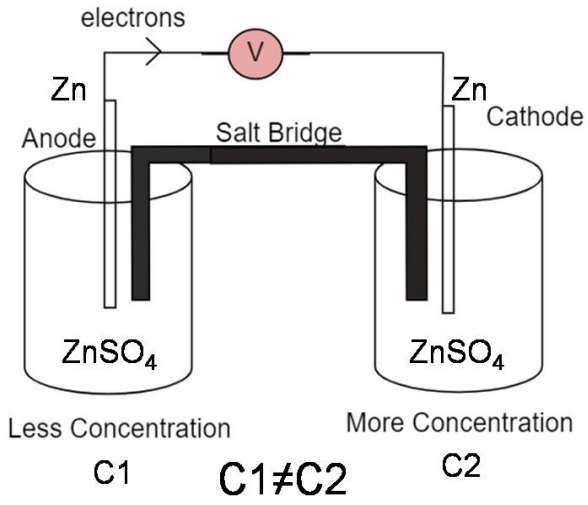

Fig. Concentration of cell

The cell is represented as $Cu/Cu^{2+}(C_1) \parallel Cu^{2+}(C_2)/Cu$

The setup of a concentration cell cosist of two electrode made with the same material dipped in the same electrolyte solution of different concentration. This type of cell is mainly used to determine pH of the unknown acid-base solution. Letus consider a conc. Cell with respect to Zn ions. The two electrode of the cell are Zn (S) I $ZnSO_4$ (C1) and $ZnSO_4$ (C2)I Zn (S)

Reprasantation of cell is : Zn (S) I $ZnSO_4$ (C1) II $ZnSO_4$ (C2)I Zn (S)

b. Working:

The working concentration of the cell depends on maintaining the equilibrium between the concentration of two half cell. Thus it creates equilibrium between the two half cells by transfer of electrons. The electrons are transferred from a half-cell having a lower concentration to half-cell having a higher concentration. The emf of cell can be found by applying Nernst equation.

The electrode reactions are given by:

Anode: $Zn(s) \rightarrow Zn^{2+}(C_1) + 2e^-$

Cathode: $Zn^{2+}(C_2) + 2e^- \rightarrow Zn(s)$

Net reaction: $Zn^{2+}(C_2) \rightarrow Zn^{2+}(C_1)$

Where C_1 and C_2 are the molar concentrations of the Zn^{2+} ions in the two half-cells.

Cell potential: $E_{cell} = E_{cathode} - E_{anode}$

Electrode potential using Nernst equation:

$$E_{cathode} = E^0 + 2.303RT/nF.\log C_2 \quad \ldots (1)$$

$$E_{anode} = E^0 + 2.303RT/nF.\log C_1 \quad \ldots (2)$$

$$E_{cell} = [E^0 + 2.303RT/nF.\log C_2 - E^0 + 2.303RT/nF.\log C_1]$$

$$E_{cell} = [2.303RT/nF.\log C_2 - 2.303RT/nF.\log C_1]$$

$$\boxed{E_{cell} = 0.0591/n \times \log(C_2/C_1)}$$

Note: The following conclusions are drawn from above equation:

i) If $C_1 = C_2$, E_{cell} is zero, hence no current flows.

ii) If $C_2 > C_1$, then $\log(C_2/C_1)$ is positive and hence E is positive; thus the direction of the spontaneous reaction is from the more concentrated solution (C_2) to the less concentrated one (C_1)

iii) Higher the ratio C_2/C_1, higher is the value of cell potential.

Numerical Problems: (Please refer lab notes for more examples)

1. Calculate the potential of Daniel cell at 25^0 C, given the electrode potentials of Cu and Zn are 0.34V and –0.76V respectively.

Solution: Given- $E^0_{Zn} = -0.76V$; $E^0_{Cu} = 0.34V$

$E_{cell} = E_{cathode} – E_{anode}$

E_{cell} = 0.34 – (-0.76)

E_{cell}= 1.1 V

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2. Write the electrode reactions and Calculate the EMF of the given concentration cell at 298K, $Ag_{(s)}$ / $AgNO_3$ (0.018M) // $AgNO_3$ (1.2M) / $Ag_{(s)}$.

Solution:

At anode $Ag_{(s)} \longrightarrow Ag^{+} + e^{-}$
At cathode $Ag^{+} + e^{-} \longrightarrow Ag_{(s)}$

$$E_{cell} = \frac{0.0591}{n} \log \frac{C_2}{C_1} \text{ at 298 K}$$

$$E_{cell} = \frac{0.0591}{1} \log \frac{1.2}{0.018}$$

E_{cell} = 0.1078 V.

3. Calculate the emf of copper concentration cell at 25^0C, the copper ions ratio in the cell is 10.

Solution, Given: $$\frac{[Cu^{+2}]_{cathode}}{[Cu^{+2}]_{anode}} = \frac{C_2}{C_1} = 10$$

w.k.t $$E_{cell} = \frac{0.0591}{n} \log\left(\frac{C_2}{C_1}\right) ; \quad \text{at 298 K}$$

$$E_{cell} = \frac{0.0591}{2} \log(10)$$

E_{cell} = 0.0296 V.

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4. A concentration cell was constructed by immersing two silver electrode in 0.05M and 1M $AgNO_3$ solution. Write the cell representation, cell reactions and calculate the EMF of the cell.
Cell representation: $Ag_{(s)}|AgNO_3(0.05M) \,||\, AgNO_3(1M) \,|\, Ag_{(s)}$

At anode: $Ag(0.05M) \longrightarrow Ag^{+}(0.05M) + e^{-}$

At cathode $Ag^{+}(1M) + e^{-} \longrightarrow Ag(1M)$

Net cell reaction $Ag(0.05M) + Ag^{+} \longrightarrow Ag^{+}(0.05M) + Ag$

$$E_{cell} = \frac{0.0591}{n}\log\left(\frac{C_2}{C_1}\right)$$

$$E_{cell} = \frac{0.0591}{}\log\left(\frac{1}{0.05}\right)$$

$E_{cell} = 0.0768V$

5. The spontaneous galvanic cell Tin/Tin ion(0.024M)//Tin ion(0.064M) /Tin develop an emf of 0.0126 V at $25^{O}C$, Calculate the Valency of Tin.

$$E_{cell} = \frac{0.0591}{n}\log\left(\frac{C_2}{C_1}\right)$$

$$n = \frac{0.0591}{E_{cell}}\log\left(\frac{C_2}{C_1}\right)$$

$$n = \frac{0.0591}{0.0126}\log\left(\frac{0.064}{0.024}\right)$$

n =2(Valency of tin)

please refer the class notes for the remaining problems

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